One shared electricity-interval width does not transfer across buildings

Abstract

We test whether one absolute-residual prediction interval, calibrated on some buildings, covers hourly electricity on other buildings. On site Rat of the Building Data Genome Project 2, 24 buildings train an hour-of-week mean, 8 buildings set the interval, and 8 buildings are held out. The interval covered 139372/139372 test hours (1.0000) at a nominal coverage of 0.9000. Its width was 452.8592 kWh. The mean test load was 37.7539 kWh and the mean absolute error was 61.0987 kWh. The width is large relative to the load, so the coverage is not evidence of a useful forecast.

1. Question

Building electricity forecasts are often a single number. An interval is useful only when it is both calibrated and narrow enough for the load it describes. This paper asks whether one interval width, estimated on other buildings, transfers to held-out buildings.

2. Data

The meters are the cleaned hourly electricity file and the building metadata in Building Data Genome Project 2, release v1.0 (doi:10.5281/zenodo.3887306), described by Miller et al. (2020). The data license is CC BY 4.0. The ASHRAE Kaggle competition files were not used. A meter is kept when at least 95 percent of its hours are present. The site with the most such meters is used. At most 40 meters are kept, spread across the sorted building identifiers. Each kept building is assigned to training, calibration, or test by a fixed hash of its identifier. Hours are not shuffled across buildings.

3. Interval

The point forecast is the mean load at that hour of the week, computed on training buildings only. The interval is the inductive split-conformal interval with absolute residuals (Vovk, Gammerman and Shafer, 2005; Lei et al., 2018). The quantile uses rank ceil((n+1)*(1-alpha)) on calibration buildings, with alpha equal to 0.10. The same width is then applied to every test hour. No other conformity score is used. A language model assisted with wording; the measurements were checked against the script output.

4. Measurement

Site Rat: 24 train buildings, 8 calibration buildings, 8 test buildings.
Split-conformal interval on held-out buildings: 139372/139372 hours inside the interval (1.0000), nominal coverage 0.9000, width 452.8592 kWh.
Train-residual quantile on the same test hours: 138969/139372 (0.9971), width 270.5115 kWh.
Point-forecast MAE 61.0987 kWh. Mean test load 37.7539 kWh.
Per test building, coverage min 1.0000, median 1.0000, max 1.0000.

5. Reading the measurement

Every test hour falls inside the interval, and the same is nearly true of the quantile taken from training residuals. That coverage sits above the nominal level because one width, set by larger calibration meters, is applied to smaller test meters. Compare the width with the mean test load in the measurement section. A symmetric residual interval can also extend below zero. Weather is not an input. One site is not another site. The result is not a building-management deployment.

6. Conclusion

A nominal 90 percent electricity interval that uses one absolute residual across buildings can cover every held-out hour

and still be too wide to use. Coverage alone is the wrong claim. The file to submit states this negative finding and no stronger one.

References

Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R. J., and Wasserman, L. Distribution-free predictive inference for regression. Journal of the American Statistical Association, 113(523), 1094-1111, 2018.

Miller, C., et al. The Building Data Genome Project 2. Scientific Data, 7, 368, 2020. doi:10.1038/s41597-020-00712-x.

Vovk, V., Gammerman, A., and Shafer, G. Algorithmic Learning in a Random World. Springer, 2005.

Dataset archive: doi:10.5281/zenodo.3887306. License: CC BY 4.0.